pdp11_34_power_sop_v1.json

```json
{
    "id": "SOP-PDP11-001",
    "title": "PDP-11/34 Power Supply (H777) Diagnostic & Troubleshooting",
    "system": "PDP-11/34",
    "component": "H777 Power Supply",
    "author": "SymbioStreams Semantic Bridge",
    "source": "Bitsavers (EK-KD11E-TM-001)",
    "steps": [
        {
            "step": 1,
            "action": "Visual Inspection",
            "details": "Check the rear circuit breaker and ensure it is in the ON position. Check for blown fuses on the power supply chassis.",
            "indicator": "Breaker tripped or fuse wire broken."
        },
        {
            "step": 2,
            "action": "Voltage Rails Measurement",
            "details": "Using a voltmeter, check the DC output voltages at the backplane connectors. Target voltages: +5V, -15V, +15V.",
            "tolerances": {
                "+5V": "4.85V to 5.25V",
                "-15V": "-14.25V to -15.75V",
                "+15V": "14.25V to 15.75V"
            }
        },
        {
            "step": 3,
            "action": "Check AC LO and DC LO Signals",
            "details": "Verify the status of AC LO and DC LO signals on the backplane. These signals must be high (>3V) for the processor to run.",
            "indicator": "If low, the power supply is signaling a power failure state."
        },
        {
            "step": 4,
            "action": "Load Isolation",
            "details": "If voltages are low, power off and remove all system modules (CPU, Memory). Power on and re-measure.",
            "inference": "If voltages return to normal, a specific module is drawing excessive current (short circuit)."
        }
    ],
    "diagnostic_flowchart": [
        {
            "condition": "No power light",
            "next": "Check AC source and fuse"
        },
        {
            "condition": "Fan running but no boot",
            "next": "Measure +5V rail"
        },
        {
            "condition": "+5V rail < 4.5V",
            "next": "Adjust +5V regulator or check for overload"
        }
    ]
}
```